Quantitative Investing

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Quant Equity Strategies

- Quantitative equity investment
  - Following **automated trading algorithms**
  - Often **small alpha** per trade, but many positions
  - Remember the Fundamental Law of Asset Management

$$S_P^2 = S_m^2 + \sum_{i=1}^{N} \left[\frac{\alpha_i}{\sigma\left(\varepsilon_i\right)} \right]^2 \tag{1}$$

- **Advantages** of quant strategies relative to discretionary strategies
  - Many stocks: **diversification**

  - **Free from behavioral biases** that afflict humans
  - You can do **backtesting**

- **Disadvantages** of quant strategies
  - **Hard** to tailor the strategy to **specific events** (e.g. news about a company)
  - **Hard** to use **soft information** such as what you get from phone calls with management – although **AI technology** can help here

Three types of quant strategies

1. **Fundamental quant**
   - Turnover measured in weeks or months
   - Trading on factors and styles
2. **Statistical arbitrage**
   - Turnover measured in hours or days
   - Trading on temporary mispricing between pairs of securities

3. **High-frequency trading**

- Turnover measured in micro-seconds
- Trading on temporary "true" arbitrages, temporary mispricings, and liquidity provision

Fundamental Quant

Fundamental Quant

- Trade on quantitative factors such as
  - Value
  - Momentum
  - Quality
  - Betting-against-beta and low volatility
  - Many others: Factor Zoo

- Use information **similar to** that used by a **discretionary equity manager**
  - Try to **"teach" a computer** what a good equity analysts does
  - Apply this across **thousands of stocks** around the world
  - In a systematic manner: **pre-specified signals and trading rules**

- Hedge funds go long and short to eliminate several sources of risk
  - **What risk is left?** Hint: AQR has been underperforming for several years

Value-Based Stock Selection

Value investing compares the current market price with some fundamental anchor to deem whether an asset is cheap or rich.

- In a typical implementation, you sort stock by some measure of fundamental value to price, and go **long** or overweight the cheap (**value**) stocks, while **shorting** or underwighting expensive (**growh**) stocks

- Fama and French introduced **book value** as fundamental anchor. Then, stocks are sorted based on the book-to-market ratio (B/M or B/P) and a factor is built going long in high B/M and short in low B/M stocks (**HML**)

Factor Construction

- However, book value is a very **coarse proxy** for fundamental value, **especially for growing companies**

- Many value metrics exist beyond the book-to-market ratio
  - Earning/P, Dividend/P, Cashflow/P, Sales/P, **Composite** of these measures
  - **Intrinsic Value-to-Price** (Bergen, Franzoni, Obrycki, Resendes, 2025, FAJ)
  - **Reversals**. E.g., prior-five-year returns

- Many ways to trade on each value metric
  - **Universe**: large, small, all stocks. Value works better with small stocks

- **Peer group**: sort across all stocks vs. intra-industry
- **Rebalancing rule**: Annual (Fama-French) or quarterly, or monthly
- **Timing of the signal**: prior year (as in Fama and French) vs. the most recent value (Asness and Frazzini, 2013)

Economic Rationale: Overreaction or Risk?

1. **Behavioral theories**
   - People overreact to news or over-extrapolate past returns
   - When investors realize about their mistakes, prices move back towards fundamental

2. **Rational risk premium**
   - Value stocks are more exposed to financial distress during bad times

- In either case, the prediction is that value stocks should pay higher returns than growth stocks

More on Performance

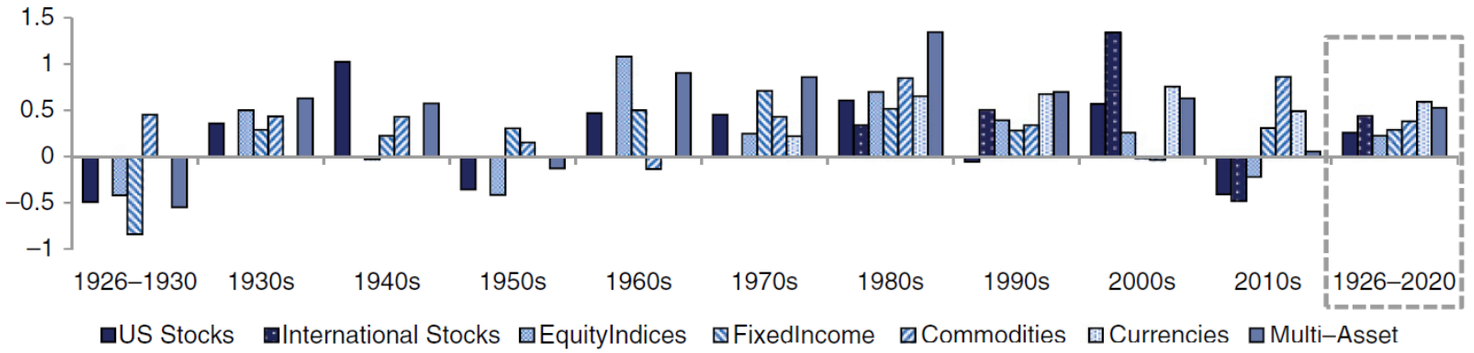
- **Small > Large**
- **Foreign > US**
- **Composite > B/P**
- **Industry Adj. > Raw**

Table 6.2 Long-Run Sharpe Ratios for Value-Based Stock Selection Strategies, 1984–Jun 2021

	US LargeCap	US SmallCap	Ex-US LargeCap	Ex-US SmallCap
BP Raw	0.15	0.16	0.35	0.30
Multi Raw	0.31	0.39	0.42	0.64
BP IndAdj.	0.28	0.32	0.65	0.43
Multi IndAdj.	0.50	0.79	0.73	0.99

Good Performance Across Asset Classes

- **Fixed income**: based on real yield
- **Commodities**: based on reversals
- **Currencies**: based on real exchange rate. It should be 1 according to purchasing-power-parity



Risks in Value Strategies

- Value goes through **protracted periods of underperformance**. Consistent with underlying risk factor
- **Worst drawdown: 63%, 2007-2020**
- Other ones: 59%, 1890-1904 (end of period of monoply of large firms); 53%, 1926-32 (Great Depression); 35% (Tech Bubble)
- Value is **short structural changes**:
  - Investors overestimate new technologies in the short run. Bad for value

Current Outlook

- Many investors have abandoned value strategies after the recent underperformance
- But recent outperformance of growth stocks is not justified by great improvement in fundamentals
- So, **opportunities never been so good** for Value (This is AQR's pitch)
- The next chart plots the **Value Spread**: **ratio of B/P of Value to B/P of Growth stocks**
  - It captures undervaluation of Value and it is used as **signal to time the value factor**

Momentum and Other Extrapolative Strategies

Extrapolative strategies buy recent winners and sell recent losers

- On **single assets** using only their past history: **directional trend following** (**CTA hedge funds**)
  - Buying/selling futures on index if performance is positive/negative
- Or, on a **cross-section of assets** using long/short portfolios based on relative performance: **momentum**
- Also called: **Absolute (or time-series) vs. Relative (or cross-sectional) momentum**

- **Cross-sectional: close to market neutral**

- **Time-series: highly directional**

Constructing the strategies

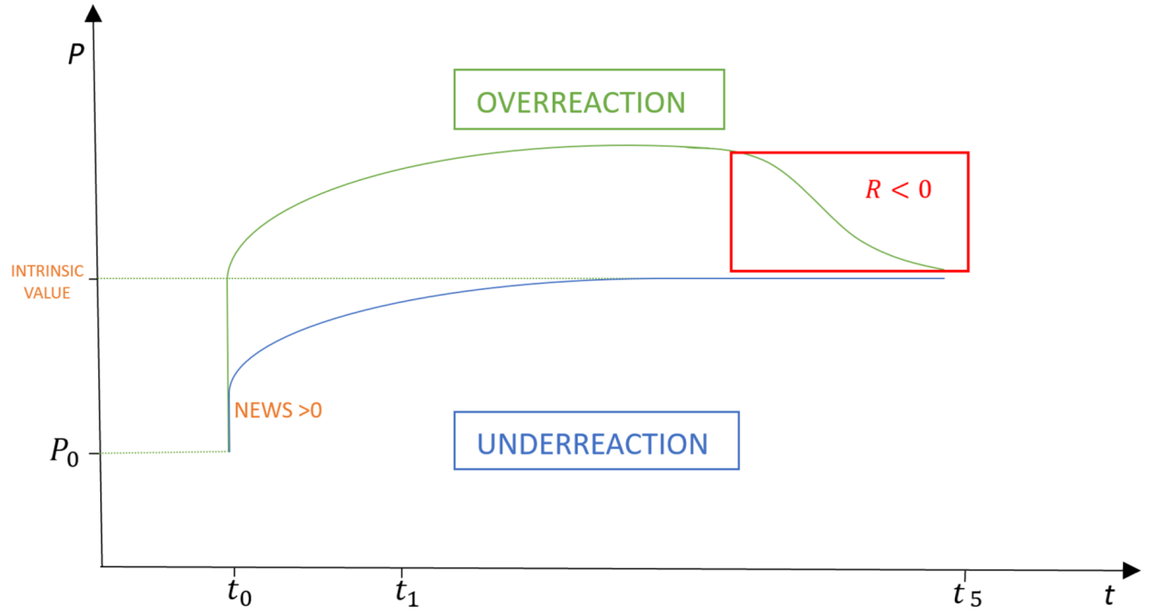
- Cross-sectional momentum
  - Compute the return of each stock for the last J months ($J = 3, 6, 9, 12$)
  - Skip the latest month because of short-term reversal (see later)
  - Buy Winners (top 10%). Sell Losers (bottom 10%)
  - Rebalance monthly
- Trend following
  - Buy assets with positive trend over the last J months ($J = 3, 6, 9, 12$)

- In forming a portfolio of assets, use inverse volatility weighting (as in risk parity)

Economic Rationale

- **Risk-based** (Rational) explanation:
  - High-momentum stocks would be more risky because they have **more growth options in earnings** and are exposed to shocks
  - Hard to believe, because high turnover of momentum strategy would imply frequent changes in the risk characteristics of a stock
- **Behavioral**: Initial **under-reaction** to news
  - Information flows slowly into prices
  - Good news today leads to a price increase today – but, price move is too small, so price must continue to go up tomorrow

- **Behavioral**: Delayed **over-reaction**
  - Good news leads to a price increase today, tomorrow people jump on the bandwagon and the price goes up further
  - Long-run reversal is a symptom of overreaction



Pitfalls, Advantages, and Outlook

- For cross-sectional momentum: **sharp market turns may lead to sudden crashes**
  - Summer 1932 (junk stocks' rally); Spring 2009 (end of bear market); November 2020 (Covid vaccines approved)
  - During the bear market, you are long low-beta stock (winners) and short high beta stocks (losers). When the market turns, you are shorting the good stocks and long the less-good stocks
- **For trend-following**: the problem is when the market is **whipsawing** (ups and downs). There is no clear trend to follow
- **Trend following** has provided a **hedge against market downturns** (see class on risk management)

  - Especially, if the downturn progresses slowly and is **long**

- The **recent underperformance** of trend following can be explained by central bank policies that target stability in financial conditions (the **"Fed Put"**)
  - They do not let markets grow/fall too much

Reconciling Cross-Sectional Momentum and Reversal

- **Short-term reversal**
  - Return over past **1 month**
  - Buy losers, sell winners
  - Possible explanation: demand pressure and **liquidity effects**
- **Momentum**
  - Return over past **12 months**, excluding the last month
  - Buy winners, sell losers

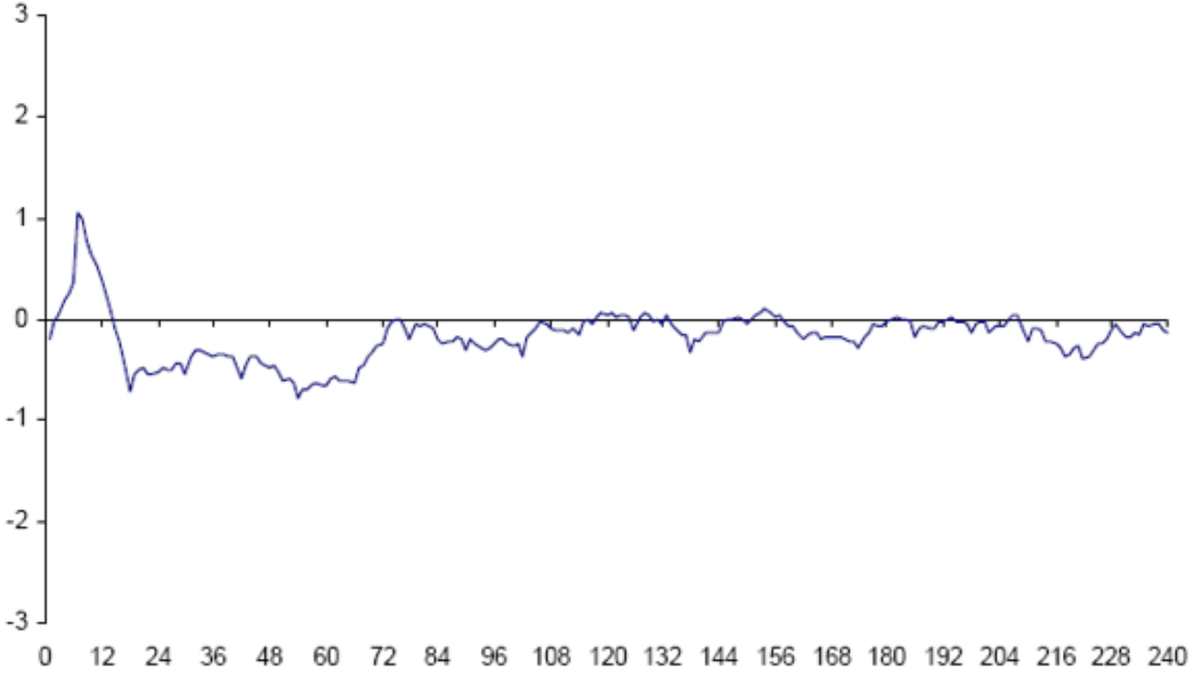
  - Possible explanations: **underreaction or delayed overreaction**

- **Long-run reversal**
  - Return over past **60 months** excluding the last 12 months
  - Buy losers, sell winners
  - Possible explanation: **overreaction**

- The three effects are summarized in the following chart reporting the estimate $\hat{\gamma}_k$ from the regression

$$R_t = \alpha_k + \gamma_k R_{t-k} + \varepsilon_t$$

Source: Heston and Sadka



Panel B: Smoothed graph

Figure 4. Cross-sectional regressions of return. Monthly cross-sectional univariate regressions of the form $x_{i,t} = \alpha_{k,t} + \gamma_{k,t}x_{i,t-k} + u_{i,t}$ are calculated for each month t and lag k, and where $x_{i,t}$ is return of stock i in month t. The lagged variable $x_{i,t-k}$ is return of stock i in month $t–k$. The regression is calculated for every month t from January 1965 through December 2002 (456 months), and for lag k values 1 through 240. Panel A plots the time-series averages of $\gamma_{k,t}$. Panel B presents a smoothed version of Panel A by replacing each observation in Panel A with the average return using a moving window of [–5,+5]. All regression estimates are reported in percent. The analysis uses NYSE/AMEX-listed stocks.

- A stock with favorable value and momentum characteristics is **a cheap stock on the rise**
  - More **likely to continue rising** than other winners, because it is still cheap
  - **Unlikely to be a value trap** because price is rising

- Hence, **momentum** can be considered as a **catalyst for value stocks**

Quality Investing

- Buy high-quality stocks, short low-quality stocks

- High-quality:
  - **Profitable, growing, stable, and well managed** (see class on Discretionary investing)

- The most typical quality signal is **profitability**, defined as

$$ROA = \frac{Net\ Income}{Total\ Assets}$$

- Comparison with Value Investing
  - **Quality** investing **buys "good" stocks** and **shorts "bad" stocks**
  - In contrast, **Value shorts the expensive stocks** (whether or not they deserve to be expensive) and **buys the cheap stocks** (whether or not they deserve to be cheap)
- Hence, **Quality complements Value**
- The two strategies are **negatively correlated**
- Combining quality and value: **Quality at a reasonable price**
- **Combining quality, value, and momentum** is even better: **buy cheap, good-quality stocks, whose price is trending up** – i.e., they have a catalyst: **SR=0.80**

Betting against Beta

- According to CAPM, the following Security Market Line should hold

$$E(R_i) - R_f = \beta_i [E(R_M) - R_f]$$

- However, the CAPM does not hold empirically as the **average return of high-beta stocks is very similar to the average return of low-beta stocks**

- In other words, the **Security Market Line is too flat empirically**

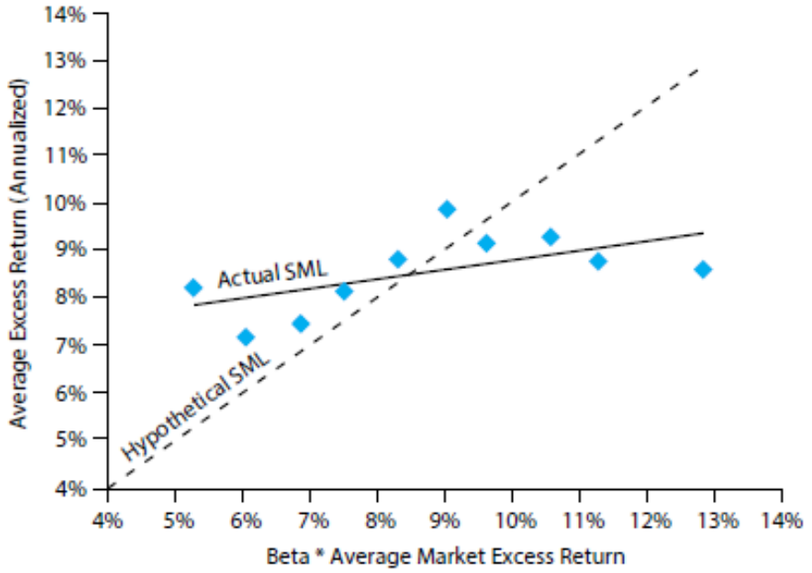


Figure 9.3. The security market line is too flat relative to the CAPM.
The figure plots ten dots corresponding to ten U.S. stock portfolios sorted by their ex ante beta, 1926 to 2010. The horizontal axis shows the CAPM-predicted return for each portfolio, i.e., its ex post realized beta multiplied by the market risk premium, $\beta^i \mathrm{E}(r_t^M - r^f)$. The vertical axis plots each portfolio's actual average excess returns, corresponding to $\mathrm{E}(r_t^i - r^f)$. The 45-degree line is the hypothetical SML implied by the CAPM.

- As a result, the **low-beta stocks have positive alpha** and vice versa

- How do you construct a market neutral portfolio that exploits these alphas?

- If you simply go long $\$1$ of low-beta stocks and short $\$1$ of high-beta stocks, you do not obtain a positive spread, because the two portfolios have similar returns
  - Also, the portfolio is not market neutral because the the long and short legs have different betas

- Instead, you can **lever up the low-beta stocks and lever down the high-beta stocks dividing by their own betas** and you obtain the betting-against-beta (**BAB**) factor of Frazzini and Pedersen (2014)

$$BAB = \frac{1}{\beta_L}(R_L - R_f) - \frac{1}{\beta_H}(R_H - R_f) \tag{2}$$

  where R_L and R_H are *characteristic weighted portfolios* (see appendix for analytical description) of low- and high-beta stocks, respectively

- In equation (2), you have that $\frac{1}{\beta_L}$ is 1.4 and $\frac{1}{\beta_H}$ is 0.7. So, the average return of BAB is positive because $R_L \approx R_H$

- The BAB portfolio is **beta neutral** by construction, but it is **not dollar neutral**
  - It goes long $\$1.4$ and short $\$0.7$. The difference coming from borrowed money (leverage)
- The SR of BAB is **0.74**

Economic Rationale for BAB Performance

- Many investors face **leverage constraints** (e.g. mutual funds)
- Thus, investors looking to pick up a higher return buy high-beta securities
- This behavior pushes prices up and expected returns down of risky assets
- Hence, the **Security Market Lines is flatter**
- **Supporting evidence** for this story
  - Mutual funds and retail investors hold stocks with betas>1 on average

  - Instead, LBO funds and Warren Buffett hold low-beta assets and lever them up (as in a BAB strategy)

- Another version of this strategy buys stocks with low volatility, which have positive alpha: **Low Volatility Anomaly**

The Quant Meltdown of August 2007

- Quant strategies are exposed to **extreme events**
- In **August 2007**, large crash in the quant space
  - Hidden from outside observers
  - Indeed, the stock market was doing well
- June/July 2007: some funds suffered losses from subprime mortgages
- Some large funds started to **deleverage** their portfolios

- The trading strategies were **crowded**, i.e. large overlapping in portfolios across different funds

- Thus, the **losses propagated** to other funds running similar strategies
  - Very **large losses** for quant funds over a **few days** in early August
  - Then, recovery